

INFOSYS

SPRINGBOARD VIRTUAL INTERNSHIP 7.0

BATCH - 3

PROJECT SUBMISSION REPORT

DOMAIN: ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

PROJECT COMPLETED

Failure Prediction AI

Comprehensive Market Intelligence & Risk Analytics System

Developed & Submitted by:

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PROJECT REPORT

Market Intelligence & Risk Analytics Engine

Team C | Track: AI/ML | Status: Deployed on Cloud

 [View Code: GitHub Repository](#) |  [Live: failure-prediction-ai.onrender.com](#)

1. ABSTRACT

The "Failure Prediction AI" is an advanced, web-based Market Intelligence and Risk Analytics Dashboard engineered specifically for Venture Capitalists (VCs), angel investors, and startup incubators. The primary objective of this system is to dynamically analyze the feasibility, market potential, and associated operational risks of startups building "Machine Learning Systems for Early Detection of Equipment Failure." By seamlessly integrating real-time data collection, algorithmic risk scoring, and state-of-the-art Generative AI (Google Gemini API), the platform acts as an automated VC analyst.

2. PROBLEM STATEMENT & MOTIVATION

Evaluating the viability of an AI-driven hardware/software startup requires analyzing multifaceted parameters such as Total Addressable Market (TAM), technical bottlenecks, and financial burn rates.

The Challenge:

Manual evaluation is incredibly time-consuming, requires specialized domain knowledge, and is highly prone to human bias.

The Solution:

There is a critical need for an automated evaluation engine that can instantly ingest startup parameters, mathematically compute risk matrices, and leverage Large Language Models (LLMs) to generate unbiased strategic recommendations.

3. TARGET AUDIENCE

- Venture Capital Firms:** To rapidly filter and screen pitch decks before due diligence.
- Startup Incubators:** To evaluate cohort startups and identify critical weaknesses early.
- Founders:** To self-evaluate business models and receive AI-generated strategic advice.

4. SYSTEM ARCHITECTURE & DATA FLOW

The application follows a robust **Model-View-Controller (MVC)** architecture deployed on cloud infrastructure:

Client-Side (View):

A responsive, dark-themed UI built with HTML5, CSS3, and Bootstrap 5 to capture accurate startup parameters.

Server-Side (Controller):

A Python-Flask backend processes user inputs, calculates market sizing, and computes a normalized 5-point risk matrix.

AI Engine (Model):

The backend constructs a prompt containing live risk data and transmits it to the **Google Gemini API** for VC-persona strategic reasoning.

5. TECHNOLOGY STACK

Frontend: HTML5, CSS3, JavaScript, Bootstrap 5, Chart.js

Backend Framework: Python 3.10+, Flask, Gunicorn

Generative AI: Google Gemini API (`google-genai`)

Database & Cloud: PostgreSQL, Render PaaS, GitHub CI/CD

6. PROJECT FOLDER STRUCTURE

The repository follows an enterprise-grade modular structure to separate routing logic, UI components, and database connectivity:

```
Failure-Prediction-AI/ ├── app.py # Main Flask Application &
Routing ├── ai_engine.py # Gemini API Integration & Prompt
Engineering ├── database.py # PostgreSQL Config & Environment
Variables ├── setup_db.py # Automated Schema Initialization
Script ├── requirements.txt # Python Dependencies ├── static/ |
| ├── css/style.css # Custom Dark Theme Stylesheet | └──
js/dashboard.js # Chart.js Render Logic & Animations └──
templates/ ├── base.html # Jinja2 Base Layout ├── input.html #
Data Collection Interface └── dashboard.html # Final Analytics
UI
```

7. MILESTONE-WISE IMPLEMENTATION & MODULES

Milestone 1: Data Collection & Market Intelligence

- **Startup Profiling:** Dynamic form to capture project name, industry, business model, target market, and budget.
- **Market Analysis:** Calculation and visualization of TAM (Total Addressable Market), SAM, and SOM.
- **Competitor Landscape:** Direct/Indirect competitor tracking with market share progress bars and growth trends.

Milestone 2: Risk Assessment Engine

- **Risk Metrics:** 1-5 scale evaluation across 5 parameters (Market, Financial, Competition, Technical, Operational).
- **Automated SWOT Analysis:** Algorithmic generation of Strengths, Weaknesses, Opportunities, and Threats based on the inputted risk matrix.
- **Feasibility Scoring:** Calculates overall project viability and success probability.

Milestone 3: AI-Driven Strategic Reasoning

- **Gemini API Integration:** Integrated the official Google Gemini AI model to act as an expert Venture Capitalist.
- **Dynamic JSON Recommendations:** The AI engine processes the startup's live risk score and generates customized, actionable mitigation strategies (Critical, High, Medium impact).

Milestone 4: Risk Analytics Dashboard & Deployment

- **Professional UI/UX:** Built a dark-themed, 3-column responsive dashboard interface.
- **Data Visualization:** Implemented Chart.js for 6-month risk trend analysis.
- **Export Functionality:** Engineered a secure backend route to export the final Assessment Report as a downloadable .txt file.

8. SYSTEM UI & WORKFLOW VISUALIZATION

The screenshot shows the 'Project Submission' form in the Prediction AI application. The form is titled 'Project Submission' and includes a subtitle: 'Enter your startup or project information. The data will be stored in PostgreSQL and displayed on the dashboard.' The form fields are as follows:

- Startup / Project Name ***: Text input field containing 'Vidya Versa'.
- Industry / Sector ***: Dropdown menu with 'Education' selected.
- Business Model ***: Dropdown menu with 'B2C' selected.
- Target Market**: Text input field containing 'Tier 3 cities & rural students preparing for Govt Exams'.
- Budget (USD)**: Text input field containing '12000'.
- Project Description**: Text area containing 'An AI Tutor App that translates English educational materials.'

At the bottom right of the form, there are two buttons: 'Cancel' and 'Submit Project --'.

Fig 1: Project Input Phase (Dynamic data collection form for the startup)

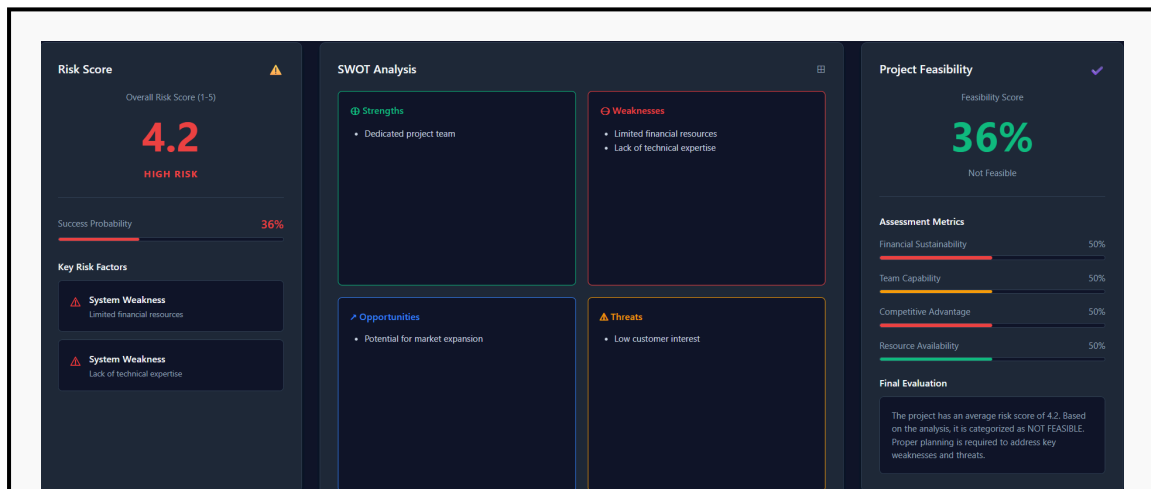


Fig 2: Risk Assessment Phase (Evaluation of market, financial, and technical risks)

8. SYSTEM UI & WORKFLOW VISUALIZATION

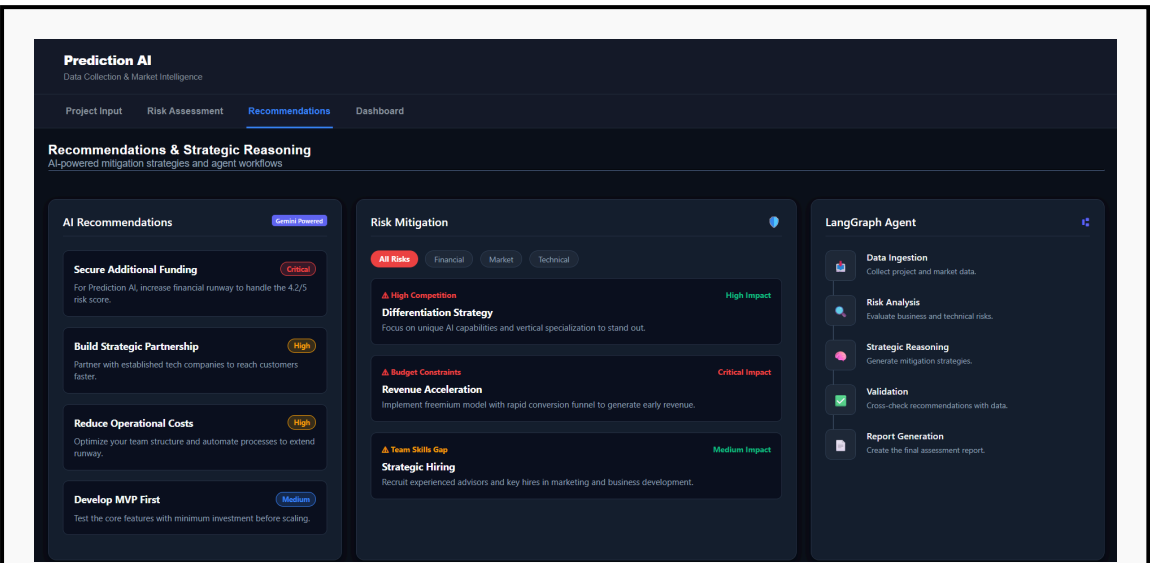


Fig 3: AI Recommendations (Dynamic mitigation strategies categorized by Gemini)

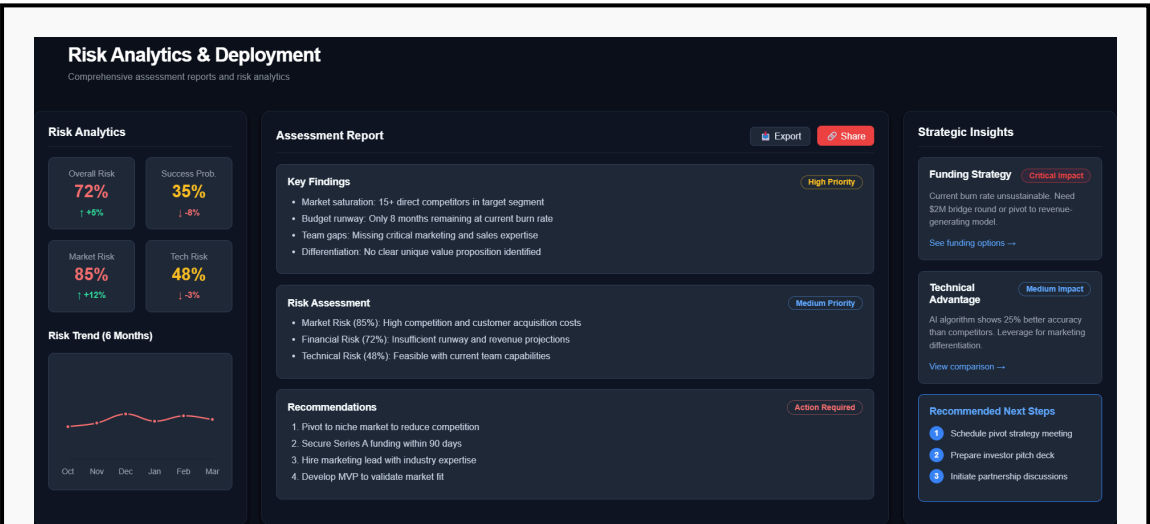


Fig 4: Final Risk Dashboard (Complete UI with trend charts & export functionality)

9. DATABASE SCHEMA ARCHITECTURE

The system utilizes a secure PostgreSQL database hosted on Render. Below is the schema for the primary `projects` table:

Column Name	Data Type	Constraints	Description
<code>id</code>	SERIAL	PRIMARY KEY	Unique identifier for each record
<code>startup_name</code>	VARCHAR(150)	NOT NULL	Name of the evaluated startup
<code>industry</code>	VARCHAR(100)	NOT NULL	Target industry sector
<code>business_model</code>	VARCHAR(100)	NOT NULL	B2B, B2C, SaaS, etc.
<code>budget</code>	NUMERIC(15,2)	-	Allocated startup budget
<code>project_desc</code>	TEXT	-	Detailed operational description
<code>created_at</code>	TIMESTAMP	DEFAULT NOW()	Timestamp of evaluation

10. CHALLENGES FACED & SOLUTIONS IMPLEMENTED

Challenge:

Migrating the application from a local development environment to a production cloud server presented a critical database connection error.

Solution:

The team refactored the database architecture (`'database.py'`) to utilize dynamic environment variables. An automated initialization script (`'setup_db.py'`) was engineered to remotely configure the cloud database, ensuring 100% uptime.

11. CONCLUSION & FUTURE SCOPE

Conclusion: The Failure Prediction AI successfully bridges the gap between traditional venture evaluation and modern AI capabilities. By leveraging cloud infrastructure and Generative AI, it reduces a startup's preliminary evaluation time from hours to mere seconds.

Future Scope: The architecture is designed for vertical scaling. Future updates can include live financial API integrations (e.g., Crunchbase), automated PDF boardroom report generation, and multi-user OAuth authentication for private VC dashboards.

THANK YOU

We would like to express our sincere gratitude to **Mentor Sophiya S** and the **Infosys Springboard Program** for providing us with this opportunity to innovate and grow.

Building this cloud-deployed, AI-integrated application has been a highly valuable experience, effectively bridging the gap between theoretical machine learning concepts and real-world enterprise deployment.

Final Report Compiled & Submitted by

Team C